Gan Boon Chai Janson

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LinkedIn | GitHub

Career Summary

Versatile software developer with experience building full-stack applications using React, HTML, CSS, JavaScript, Node.js, and PostgreSQL. Contributed to the mySHAREJourney web and mobile application by delivering responsive, user-friendly features that improved accessibility and engagement. Proficient in Python and SQL for developing data workflows that enhance application insights and decision-making. Detail-oriented and collaborative professional seeking a role that combines software engineering and data engineering to build data-informed applications that drive efficiency and support key business outcomes.

Professional Development

Generation Singapore (In Partnership with Microsoft and Temasek Polytechnic)

Junior Data Engineer Trainee **Nov 2025 – Feb 2026**

- Built and validated data pipelines using Python, SQL, and PySpark, where attention to detail was critical for ensuring data accuracy and completeness in every transformation step.
- Designed database schemas and performed ETL tasks by collaborating with teammates to translate business requirements into technical solutions, ensuring project milestones were met on time.
- Developed data models and visual insights using Power BI, enhancing team communication and decision-making through clear, data-driven dashboards.
- Applied structured problem-solving techniques to debug pipeline issues and optimize data workflows, demonstrating personal responsibility and resilience when tackling complex technical challenges.

Capstone Project: Singapore HDB Resales ETL Project (Joint Project)

- Task: Load Phase
- Objectives: Built an end-to-end ETL pipeline to process Singapore HDB resale flat data. Delivered a reliable PostgreSQL database containing clean, structured resale data for analysis and reporting.
- Designed and implemented the Load process to insert transformed data into PostgreSQL.
- Defined and maintained database schema using SQLAlchemy for consistency and scalability.
- Automated table creation and schema validation to reduce manual setup errors.
- Implemented batch loading (5,000 rows per chunk) to optimize performance and handle large datasets efficiently.
- Added logging and error handling to ensure traceability and reliability of data loads.

NTUC LearningHub

Desktop Support Engineer Trainee **July 2025 – Oct 2025**

- Completed hands-on training in hardware/software troubleshooting and core networking (TCP/IP, Wi-Fi, SOHO), supporting simulated resolution of 20+ common IT issues.
- Gained practical experience in Microsoft 365 Endpoint Administration and Apple device support, managing device policies and security in lab-based environments.

- Applied the FIRE framework to identify and propose improvements to 2 workplace processes through structured problem-solving sessions.
- Used the CENTRIC framework to develop a customer-focused tech solution prototype based on user data and iterative feedback cycles.

Capstone Project: Network deployment, VM deployment, and troubleshooting.

Career Break **June 2024 – Oct 2025**

- Prioritized health and wellness to ensure readiness for a career transition.
- Completed structured online courses in Python and SQL, building a strong foundation in programming for data manipulation.
- Applied learning by creating a portfolio of Python mini-projects that demonstrate practical problem-solving and coding ability.
- Completed a 3 months Desktop Support Engineer bootcamp, gaining skills in hardware, software, and network troubleshooting to build a comprehensive IT foundation

Work Experience

Activate Interactive Pte Ltd

Programmer Analyst **Sept 2022 - May 2024**

- Develop full-stack development of the "mySHAREJourney" application, architecting the React JS & React Native frontend and building the supporting Node.js & Express.js REST API with a PostgreSQL database.
- Translated business requirements into technical solutions, ensuring alignment with stakeholder needs and improving feature delivery accuracy by 15%.
- Collaborated with cross-functional teams (designers, QA, product owners) in an Agile environment, successfully delivering multiple feature iterations within sprint timelines.
- Implemented and executed automated test cases using Robot Framework, increasing test coverage and reducing manual testing effort by 30%.
- Contributed to all phases of the SDLC — from requirements analysis to coding, testing, deployment, and post-release support — gaining comprehensive experience in enterprise-grade application development.

Activate Interactive Pte Ltd

Software Developer Intern **April 2022 – Sept 2022**

- Gained in-depth knowledge of React Native and microservices architecture through mentorship from senior developers, strengthening understanding of scalable mobile and backend systems.
- Engaged in Agile ceremonies (daily stand-ups, sprint planning, retrospectives), developing familiarity with the full software development lifecycle and collaborative project management.
- Enhanced technical foundation by exploring modern development practices such as modular service design, API integration, and cross-platform mobile development.

Generation Singapore (In Partnership with Microsoft and Temasek Polytechnic)

Junior Full Stack Developer Trainee **Jan 2022 – April 2022**

- Built responsive, component-based user interfaces using HTML5, CSS3, JavaScript (ES6+), and React, focusing on modern frontend development principles

- Built back-end functionality and managed database queries with SQL, practicing personal responsibility in debugging issues and optimizing queries for performance.
- Managed the full project lifecycle using Git and GitHub in a collaborative Agile team, strengthening teamwork and communication through daily stand-ups, peer code reviews, and sprint planning.
- Applied structured time management to deliver individual tasks within sprint deadlines, demonstrating resilience and a growth mindset when troubleshooting complex technical challenges during development.

Capstone Project: Web Development Portfolio

- As part of a collaborative full-stack development team, I was the front-end developer responsible for building and styling the responsive homepage (index.html) for a business website.
- Developed the user interface using HTML, CSS, and JavaScript, ensuring a clean, accessible, and fully responsive design that adapts to different screen sizes.
- Implemented the core content sections, including the business introduction ("Our Shop") and key service listings ("Most Preferred Services")—focusing on clear information architecture and visual hierarchy.

SMRT Buses Ltd

Bus Simulator Driving Trainer **Aug 2017 – Oct 2021**

- Designed and delivered defensive driving curriculum using Unity-based driving simulator scenarios, training 200+ Bus Captains and improving safety awareness scores in post-training evaluations.
- Developed interactive practice modules that reduced simulation errors by 25%, boosting Bus Captain confidence and performance during driving assessments.
- Collaborated with engineers to troubleshoot and resolve 10+ recurring iDSC system issues, increasing simulator uptime to 95% and ensuring smoother training sessions.
- Digitized 3 years of paper-based training attendance records into a structured database using PHP, MySQL(via phpMyAdmin), and built a web-based dashboard (HTML, JavaScript, CSS) to visualize statistics — reducing manual tracking time by 50% and enabling real-time performance insights.

Projects: Singapore 2019 ITS World Congress, Smart Nation & U Exhibition **Oct 2019 – Dec 2019**

- Conducted daily testing of Autonomous Vehicle (AV) systems, ensuring responsiveness to simulated threats and safety scenarios.
- Evaluated AV performance metrics during testing and provided structured feedback to research teams and engineers, contributing to iterative improvements.
- Demonstrated AV capabilities to 500+ exhibition visitors, showcasing smart mobility innovations and supporting public engagement with emerging technologies.

Singapore Technologies Kinetics Ltd

Driving Trainer **Mar 2012 – July 2017**

- Educated 500+ NSF trainees on the structure and features of Class 3 and Class 4 military vehicles, improving technical understanding of vehicle components and operational readiness.
- Delivered defensive driving and traffic rule training, reducing accident risk during training exercises and strengthening safety culture among new drivers.

- Trained recruits in fundamentals of driving across both road and off-road terrain, increasing trainee confidence scores in evaluations and ensuring operational competence in diverse environments.
- Maintained and monitored training equipment and resources, ensuring 100% availability during scheduled sessions and minimizing downtime.

Singapore Bukit Batok Driving Centre
Driving Instructor **Apr 2011 – Feb 2012**

- Conducted 300+ theoretical and practical driving lessons for motorcar learners, ensuring comprehensive coverage of road rules and vehicle operation.
- Guided and motivated 50+ assigned students to achieve their driving goals, contributing to a high pass rate in licensing examinations.
- Promoted safe and defensive driving habits, reducing training related incidents and strengthening learner confidence on both urban and highway roads.
- Maintained training equipment and resources to 100% operational standards, ensuring safety compliance and uninterrupted lesson delivery.

Early Career (Condense Summary, 1997 – 2011)

- **Sales & Customer Service**: Sales Executive at BKW Rent A Car, supporting rental operations, client relations, and contract management.
- **Hospitality**: Front Office Executive at Hotel 81, managing guest check-ins, reservations, and customer service.
- **Retail Sales**: Sales positions at Citimac Electronic, Lomax Electronic Enterprises, Mobile Storm, and Max Group companies, where responsibilities included customer engagement, product demonstrations, and achieving sales targets.
- **Transportation & Logistics**: Driver roles at Pet Movers Singapore, Mexsim Enterprise, KM Construction, and Skywave (Driver cum Technician), ensuring safe and timely delivery of goods and services.

Professional Certification

Creative Thinking (Intermediate)	Attained: July 2025
Problem Solving (Intermediate)	Attained: July 2025
AWS Certified Cloud Practitioner	Attained: Sept 2023 – Expired Sept 2026
ISTQB Foundation Level	Attained: April 2023
Implement Operations for Service Excellence	Attained: Nov 2017

Education

Singapore GCE ‘O’ Level Bendemeer Secondary School – 1991